

Testing Documentation: Periplus Shopping Cart

I. Test Case Components

A standard, well-constructed test case typically contains the following components:

Table 1.1 Test Case Components

Component	Description
Test Case ID	A unique identifier to reference the test case
Test Case Name	A short, clear statement of what is being tested.
Description	Brief explanation of test purpose and scope.
Preconditions	The required conditions before executing the test
Test Data	Specific data or inputs used in the test execution
Test Steps	A clear step-by-step sequence of user actions
Expected Result	The output expected after performing the steps
Status	Test execution outcome (Pass, Fail, Blocked, or Skipped)
Priority	Test execution order based on business priority levels (High, Medium, or Low).

To better understand how each component is used, below is a test case example that can be used as a template.

1. Sample Test Case - Verify User Login

Table 1.2 Sample Test Case Details

Component	Details
Test Case ID	TC-AUTH-001
Test Case Name	Verify user login with valid credentials
Description	Brief explanation of test purpose and scope.
Preconditions	User is already registered and has active credentials.
Test Data	Email: "user@example.com" Password: "pass123"
Status	Pass
Priority	High

Table 1.3 Sample Test Steps & Expected Results

Step	Action	Expected Result
1	Open browser and navigate to https://www.example.com/	The homepage loads completely.
2	Click on the "Sign In" button.	The browser redirects to the Login page, and the login form is displayed.
3	Enter the valid email into the Email field.	The email field accepts and displays the entered email address.
4	Enter the valid password into the Password field.	The password field accepts the input and masks the characters with dots.
5	Click the "Login" button.	The user is successfully authenticated, redirected to the Homepage, and the user's profile icon & name is visible.

II. Test Documentation (Scenario B - Shopping Cart)

To ensure the system behaves as expected, a clear and well-structured set of test documentation is needed. It helps translate requirements into actionable test cases with defined steps and expected results. The following tables outline the complete test documentation for Periplus' shopping cart.

1. Test Case 01 - Add Product to Cart as a Logged-In User

Table 2.1 TC-CART-001 Details

Component	Description
Test Case ID	TC-CART-001
Test Case Name	Verify adding a product to the cart as a logged-in user
Description	To ensure that an authenticated user can search for a product, add it to the cart, and the product is available in the shopping cart.
Preconditions	1. User has a valid registered account on Periplus 2. User has logged in
Test Data	Email: "user@example.com" Password: "pass123" Search Keyword: "Spy School Secret Service"
Status	(Tested) Pass
Priority	High

Table 2.2 TC-CART-001 Steps & Expected Results

Step	Action	Expected Result
1	Open browser and Navigate to https://www.periplus.com/	The Periplus homepage loads completely.
2	Enter the search keyword in the search bar and click the "Search" button.	The search results page loads and displays products that match the keyword.
3	Click on any product	The system navigates to the product's detail page.

4	Click the "Add to Cart" button	A successful "Add to Cart" popup is displayed on the screen.
5	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page, displays the previously added products & subtotals.

2. Test Case 02 - Add Product to Cart as a Guest

Table 2.3 TC-CART-002 Details

Component	Description
Test Case ID	TC-CART-002
Test Case Name	Verify adding a product to the cart as an unauthenticated user (Guest)
Description	To ensure that an authenticated user can search for a product, add it to the cart, and the product is available in the temporary shopping cart.
Preconditions	User is not logged in
Test Data	Search Keyword: "Spy School Secret Service"
Status	Not tested
Priority	Medium

Table 2.4 TC-CART-002 Steps & Expected Results

Step	Action	Expected Result
1	Open browser and Navigate to https://www.periplus.com/	The Periplus homepage loads completely.
2	Enter the search keyword in the search bar and click the "Search" button.	The search results page loads and displays products that match the keyword.

3	Click on any product	The system navigates to the product's detail page.
4	Click the "Add to Cart" button for a product.	The item is added to a temporary session cart, and a success notification appears.
5	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page, displays the previously added products & subtotals.

3. Test Case 03 - Checkout Product in Cart as a Logged-In User

Table 2.5 TC-CART-003 Details

Component	Description
Test Case ID	TC-CART-003
Test Case Name	Verify the checkout process for products in the cart as an authenticated user
Description	To ensure that a logged-in user can successfully proceed from the cart, input shipping and payment details, and complete the purchase.
Preconditions	1. User has logged in 2. User has at least one item inside the shopping cart.
Test Data	Shipping Address: "Jl. Ganesa No. 10 Bandung" Payment Method: "E-Wallet"
Status	Not tested
Priority	High

Table 2.6 TC-CART-003 Steps & Expected Results

Step	Action	Expected Result
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1	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page, displays the previously added products & subtotals.
2	Select products to checkout	The subtotals update automatically every time any products are selected/unselected.
3	Click the "Checkout" button.	The browser redirects to the Checkout page, displaying the user's saved payment and shipping option.
4	Select or confirm the preferred shipping address and shipping method.	The shipping cost is automatically calculated and added to the Grand Total.
5	Select a valid payment method from the available options.	The payment details section expands to show instructions for the selected method.
6	Click the "Confirm Order" button.	The system processes the transaction, pops up "success" notification, and redirects the user to the order details

4. Test Case 04 - Checkout Product in Cart as a Guest

Table 2.7 TC-CART-004 Details

Component	Description
Test Case ID	TC-CART-004
Test Case Name	Verify checkout restriction for Guest users
Description	To ensure that an unauthenticated user cannot proceed with checkout and is forced to log in or register first, without losing their cart items.
Preconditions	1. User is not logged in

	2. User has at least one item inside the shopping cart.
Test Data	Email: "user@example.com" Password: "pass123"
Status	Not tested
Priority	High

Table 2.8 TC-CART-004 Steps & Expected Results

Step	Action	Expected Result
1	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page & displays the previously added products & subtotals.
2	Select products to checkout	The subtotals update automatically every time any products are selected/unselected.
3	Click the "Checkout" button.	The system stops the checkout process and redirects to the Login page.
4	Enter valid credentials into the Login form and click the "Login" button.	The user is successfully authenticated & the system redirects the user back to the Checkout page without losing the previously added products.

5. Test Case 05 - Update Item Quantity in Cart

Table 2.9 TC-CART-005 Details

Component	Description
Test Case ID	TC-CART-005
Test Case Name	Verify that a user can update item quantity

Description	To ensure the shopping cart immediately calculates the item quantity and subtotal price when the + or - buttons are clicked.
Preconditions	1. User has exactly one item with a quantity of "1" inside the shopping cart.
Test Data	-
Status	Not tested
Priority	High

Table 2.10 TC-CART-005 Steps & Expected Results

Step	Action	Expected Result
1	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page & displays the previously added products & subtotals.
2	Click the "+" (plus) button next to the item quantity once.	The quantity number immediately changes from "1" to "2" and the subtotal price updates to its twice immediately.
3	Click the "-" (minus) button next to the item quantity once.	The quantity number immediately changes back from "2" to "1" and the subtotal price updates to its half immediately

6. Test Case 06 - Remove Item from Cart

Table 2.11 TC-CART-006Details

Component	Description
Test Case ID	TC-CART-006
Test Case Name	Verify that a user can remove an item

Description	To ensure users can delete unwanted items by first selecting the product checkbox, clicking the main trash icon at the top corner, and accepting the confirmation pop-up.
Preconditions	1. User has at least one item inside the shopping cart.
Test Data	-
Status	Not tested
Priority	High

Table 2.12 TC-CART-006 Steps & Expected Results

Step	Action	Expected Result
1	Click the "Shopping Cart" icon in the navigation bar.	The browser redirects to the shopping cart page & displays the previously added products & subtotals.
2	Click the checkbox next to any product to select it.	The checkbox is checked, and the "Trash" icon at the top corner of the application becomes visible.
3	Click the "Trash" icon located at the top corner of the application.	A deletion confirmation pop-up appears on the screen.
4	Click the "Yes" button on the confirmation pop-up.	The pop-up closes, and the selected item is immediately removed from the cart list.

III. Automated Testing on Periplus

This section covers the test automation implementation for the first test case (**TC-CART-001, Verify adding a product to the cart as a logged-in user**). The test script simulates actual user interactions on the Periplus website, from authentication to cart verification. The automated test was developed using **Java** as the programming language, **Selenium** for the web interaction, and **TestNG** for the test organization.

The automated test uses the **Page Object Model (POM) design pattern**. This approach separates the test logic from the web element locators to make sure the codes are maintainable. The test is separated into two main types of class:

- **Test Class (PeriplusCartTest.java):** Contains the actual test scenarios configured with TestNG annotations. Test Class executes the steps and handles the final Assert verifications.
- **Page Classes (HomePage.java, LoginPage.java, CartPage.java):** Represent each page of the website, containing the elements & actions related to each page. Each class stores the web element locators and methods for common user interactions.

The automated test was executed using Maven through CLI. The test successfully executes every action in the test case TC-CART-001 which shows the automated test is functioning correctly.

```
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 26.999 s - in com.periplus.carttest.PeriplusCartTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 30.621 s
[INFO] Finished at: 2026-06-12T23:06:39+07:00
[INFO] -----
```

Figure 1 Automated Test Results

The complete source code for the test automation can be accessed through the repository below.

Link: <https://github.com/usernamedarren/periplus-cart-automation-test>